Introduction to Computer Programming (ICP, CCIT4020)

HKU SPACE Community College, 2021-2022 Semester 1

(Lab 02)

(Practicing Lab, NOT for submission)

Review the related lecture note(s), and finish exercises below.

Useful Hotkeys in Python IDLE Code Editor:

- < Ctrl + **c** > = Copy (Selected "Codes")
- < Ctrl + **v** > = Paste (Copied "Codes", if any)
- < Ctrl + **x** > = Cut (Selected "Codes")
- < Ctrl + **s** > = Save the file
- < Ctrl + **z** > = Undo the last step
- < **F5** > = Run Module

1. 1a) Write and execute a Python program (named `Lab2_1a.py`) that displays the results of the expressions:

   1) `15.0 * 4.0 + 7.0`

   2) `22.2 - 4.0 * 8.5 - 3.6`

   3) `(6.5 + 4.5) * (4.0 / 2.0)`

   - Calculate the value of these expressions by typing the following programming codes to verify that the values are correct (write comments for explanation by using symbols "**#**" if necessary).
   - Display student's own name before terminating the program.

   Sample Source Code

```
# Lab2_1a.py, ICP, 2021
print ("15.0 * 4.0 + 7.0  = ", (15.0 * 4.0 + 7.0))
print ("22.2 - 4.0 * 8.5 - 3.6 = ", (22.2 - 4.0 * 8.5 - 3.6))
print ("(6.5 + 4.5) * (4.0 / 2.0) = ", (6.5 + 4.5) * (4.0 / 2.0))
print ("\n........ By YourName; ICP 4020; 2021........\n")
```

   Sample Display Output

```
15.0 * 4.0 + 7.0  =  67.0
22.2 - 4.0 * 8.5 - 3.6 =  -15.4
(6.5 + 4.5) * (4.0 / 2.0) =  22.0

........ By YourName; ICP 4020; 2021........
```

	1b) Write and execute a Python program (named `Lab2_1b.py`) that displays the results of the expressions:
	
	1) `25 * 7`
	2) `15 % 4`
	3) `6 + 7 * (10 / 5)`
	
	Sample Display Output
	
	`25 * 7 =  175`
	`15 % 4 =  3`
	`6 + 7 * (10 / 5) =  20.0`
	
	`........ By YourName; ICP 4020; 2021........`

2.	○ Write and execute a Python program that calculates and displays the sum of the numbers from `1.0` to `200.0`.
	○ The equation for calculating this sum is `sum = (n/2) (2*a + (n - 1)d)`, where `n` = number of terms to be added, `a` = the first number, and `d` = the difference between each number. (Name the program file `Lab2_2a.py`)
	
	`The sum the numbers from 1.0 to 200.0 is 20100.0`
	
	`........ By YourName; ICP 4020; 2021........`
	
	○ Modify to find the sum of the numbers from `300.0` to `1000.0`. (File `Lab2_2b.py`)
	
	`The sum the numbers from 300.0 to 1000.0 is 455650.0`
	
	`........ By YourName; ICP 4020; 2021........`

3.	○ The distance that light travels in one year is called a light year. Given that light travels at a speed of `300000000.0` meters per second, write and execute a Python program to determine the distance of a light year. (Name the program file `Lab2_3.py`)
	○ The relevant formula is `distance = speed * time`.
	○ Better create variables and use assignment statements.
	
	`The total distance light travels in one year is 9460800000000000.0 meters.`
	
	`........ By YourName; ICP 4020; 2021........`

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